

Intro To Computer Vision: Few-shot learning using Matching Networks, Prototypical Networks, and Relation Network

(Team Nutella) Github Code

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1 Introduction

Standard deep learning models for image classification have achieved remarkable performance, but they typically require large labeled datasets to generalize well. This reliance on abundant data becomes a critical bottleneck when labeled examples are scarce: a setting that arises naturally in domains such as medical imaging, rare object recognition, and personalized systems. In contrast, humans can recognize a new concept from just one or a handful of examples, highlighting a significant gap between biological and machine learning. This motivates the study of *few-shot learning*: training models that can generalize to new classes given only a small number of labeled examples per class.

Metric-based meta-learning has emerged as a dominant paradigm for few-shot classification. Rather than fine-tuning a model on each new task, these approaches learn an embedding space where class membership can be inferred by measuring distances or similarities between examples and enabling generalization to entirely unseen classes at test time. Three influential works in this space form the basis of this report: namely matching Networks [4], Prototypical Networks [2], and the Relation Network [3]. We implement each model, analyze their training dynamics, and introduce a simple inference UI.

2 Summary

Our work explored three metric-based meta-learning approaches for few-shot image classification on the Omniglot benchmark, yielding the following key outcomes:

- **Model Implementations:** We implemented and trained Matching Networks [4], Prototypical Networks [2], and Relation Networks [3] under standard N -way (number of classes per episode) K -shot (number of labeled support examples per class) episodic settings, validating each method's core contribution: attention-based nearest neighbor classification, mean-prototype embeddings, and learned relation scoring respectively.
- **Training Dynamics:** We plotted and analyzed training curves for all three models, comparing convergence behavior and generalization across shot and way configurations.
- **Interactive Inference UI:** We developed a user interface enabling few-shot inference with any of the three models without direct interaction with the underlying code.

3 Dataset

We use the Omniglot dataset [1], which consists of 1,623 handwritten characters across 50 alphabets with 20 samples per character. It is evaluated under an N -way K -shot classification setting.

4 Methodology

Here we understand and implement Matching Networks, Prototypical Networks, and Relation Network.

4.1 Matching Networks

Matching Networks [4] frame few-shot classification as a set-to-set mapping: given a support set $S = \{(x_i, y_i)\}_{i=1}^k$, the model defines a classifier $c_S(\hat{x})$ for any query $\hat{x}$ without any weight updates or fine-tuning. The prediction is a weighted sum of support labels, with weights given by a cosine-softmax attention kernel over independently embedded support and query points:

$$P(\hat{y} \mid \hat{x}, S) = \sum_{i=1}^k a(\hat{x}, x_i) y_i \quad (1)$$

$$a(\hat{x}, x_i) = \frac{\exp(c(f(\hat{x}), g(x_i)))}{\sum_j \exp(c(f(\hat{x}), g(x_j)))} \quad (2)$$

where $c(\cdot, \cdot)$ is cosine similarity and f, g are embedding functions (here $f=g$, a shared four-block CNN, whose architecture is described in 4.3.1).

Training maximises the expected log-probability of the correct label across episodically sampled tasks:

$$\theta = \arg \max_{\theta} \mathbb{E}_{L \sim \mathcal{T}} \left[\mathbb{E}_{S \sim L, B \sim L} \left[\sum_{(x,y) \in B} \log P_{\theta}(y \mid x, S) \right] \right] \quad (3)$$

where $\mathcal{T}$ is the task distribution, L is the set of classes sampled for that episode, $S \sim L$ is the support set of K labeled examples per class, and $B \sim L$ is the query batch drawn from the same classes. At each training step, L is sampled from $\mathcal{T}$, and the model is trained to correctly classify B conditioned on S , directly mirroring the N -way K -shot structure seen at test time. At inference, a new support set S' of unseen classes is presented with a new query; since the classifier is non-parametric, no retraining or fine-tuning is required. The model architecture is described in fig 1.

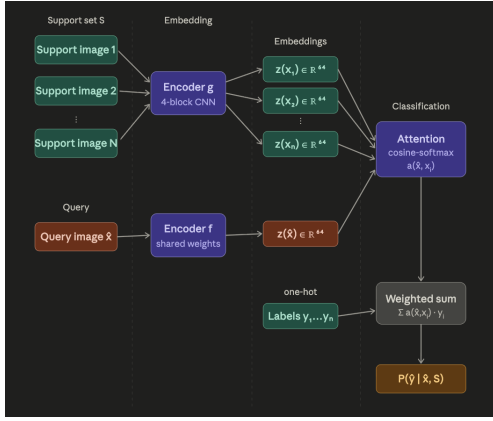


Figure 1: Matching Networks Architecture

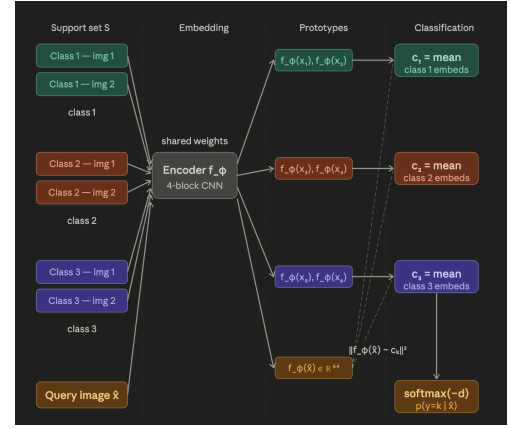


Figure 2: Prototypical Networks Architecture

4.2 Prototypical Networks

Prototypical Networks [2] learn an embedding space where each class is represented by its mean support embedding (*prototype*):

$$c_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_\phi(x_i) \quad (4)$$

where $f_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^M$ is a learned non-linear embedding function parameterised by a four-block CNN [section 4.3.1, figure 2]. A query is classified by a softmax over negative squared Euclidean distances to each prototype:

$$p_\phi(y = k | x) = \frac{\exp(-\|f_\phi(x) - c_k\|^2)}{\sum_{k'} \exp(-\|f_\phi(x) - c_{k'}\|^2)} \quad (5)$$

and training minimises $J(\phi) = -\log p_\phi(y = k | x)$ via Adam with an initial learning rate of 10^{-3} , halved every 2000 episodes. Squared Euclidean distance is chosen deliberately: as a Bregman divergence, it makes the class mean the provably optimal prototype and connects the model to exponential family mixture density estimation.

4.3 Relation Network

Here we describe the Relation Networks work.

4.3.1 Model Architecture:

The Relation Network has two modules: an **embedding module** and a **relation module**. The embedding module is a shared-weight CNN with four convolutional blocks (first two use 2×2 max-pooling, last two do not), encoding support and query images into feature maps **A** and **B** of shape $[64, H, W]$. These are concatenated depth-wise into a $[128, H, W]$ tensor and fed to the relation module: two convolutional blocks followed by two FC layers (ReLU: $64 \rightarrow 8$, Sigmoid: $8 \rightarrow 1$), producing a relation score $s \in [0, 1]$. For loss, we use MSE rather than Cross-Entropy loss.

$$\varphi, \phi \leftarrow \underset{\varphi, \phi}{\operatorname{argmin}} \sum_{i=1}^m \sum_{j=1}^n (r_{i,j} - \mathbf{1}(y_i == y_j))^2$$

4.3.2 Workflow:

Similar to Matching Networks (Section 4.1) and Prototypical Networks (Section 4.2), embeddings are generated for both support and query images. However, rather than using a fixed distance metric for classification, the embeddings are concatenated and passed to the relation module, which directly outputs a similarity score for each support-query pair.

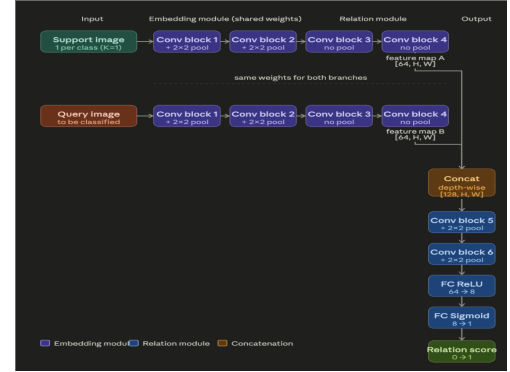


Figure 3: Relation Networks Architecture

4.3.3 Zero Shot

The paper extends the framework to zero-shot learning by replacing the support image with a semantic attribute vector, fed through a separate embedding module. The relation module then compares query image embeddings against these attribute embeddings, requiring no architectural redesign.

4.3.4 Improvement using Cross-Attention

We try to improve the system by changing the relation network architecture. Here, rather than running through a convolution layer, we run through a cross-attention layer first before passing on to the later layers. We see improvement in accuracy from 98.2% to 98.51%. We mention them as Cross Relation Net in Table 1

4.4 User Interface

We also present a Gradio based UI, as shown in Fig 4 for easy classification and visualization.

5 Results and Analysis

Here we share the results of the experiments.

Table 1: Results on Omniglot (5-way 1-shot).

Model	Train Acc. (%)	Eval Acc. (%)
Matching Nets [4]	92.64	87.37
Proto Nets [2]	99.12	98.01
Relation Nets [3]	≈ 100	98.2
Cross Relation Nets	≈ 100	98.51

- Matching Networks show a larger generalisation gap, with

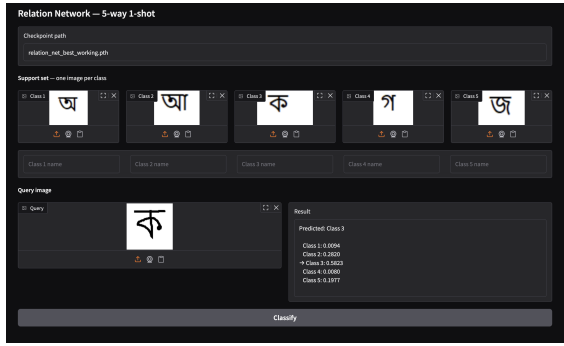


Figure 4: User Interface



Figure 5: Matching Network train and eval curves

train accuracy converging to $\sim 95\%$ while eval accuracy plateaus around 87% . The eval loss remains consistently above the train loss throughout training, suggesting the attention-based classifier overfits to the training episode distribution and does not generalise as effectively as prototype-based approaches 5.

- Prototypical Networks converge rapidly within the first few episodes, with train accuracy saturating near 99% and eval accuracy stabilising around 98% . Both train and eval loss decrease steadily and remain close, indicating minimal overfitting and strong generalisation to unseen classes 6.
- The Relation Network achieves eval accuracy exceeding 98% , with train accuracy near 100% . Both losses follow a decreasing trend, though the eval loss stays above the train loss consistent with the generalisation gap expected in few-shot settings 7.

6 Conclusion

We implemented and evaluated three metric-based meta-learning approaches - Matching Networks, Prototypical Networks, and Relation Networks - on the Omniglot 5-way 1-shot benchmark. Prototypical and Relation Networks both achieved $\sim 98\%$ eval accuracy, consistent with reported results, while Matching Networks converged to 87% . The results suggest that the choice of comparison mechanism - learned metric, Euclidean distance, or cosine similarity - has a significant impact on generalisation, with Relation and Prototypical Networks benefiting from stronger inductive biases. The results corroborate the conjecture of Snell et al. [2]: cosine similarity, used by Matching Networks, lacks the Bregman divergence property that makes the class mean the optimal prototype, and this manifests empirically as a clear drop in generalisation performance.

7 Limitations

All experiments were conducted only under the 5-way 1-shot setting; results may differ under higher-way or higher-shot configura-

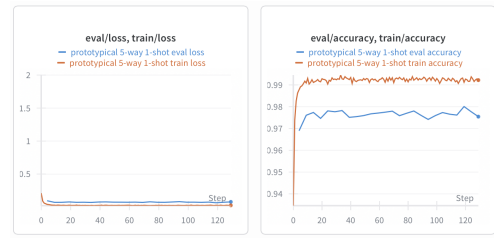


Figure 6: Prototypical Network train and eval curves

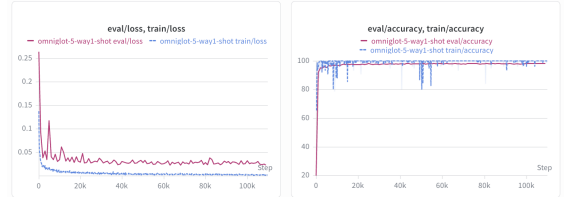


Figure 7: Relation Network train and eval curves

tions. We implement only the simple non-FCE variant of Matching Networks, where support and query points are embedded independently. The full context embedding (FCE) variant proposed by Vinyals et al. conditions each support embedding on the entire support set via a bidirectional LSTM, and the query embedding on the support set via an attention LSTM, allowing embeddings to be episode-aware rather than instance-independent. This richer contextualisation may close the performance gap with Prototypical and Relation Networks.

8 Contributions

- Riya Mahesh - Worked on summarizing (report) and training Matching Networks and Prototypical networks
- Nilarnab Debnath - Worked on summarizing (report), training Relational Networks, attention-based improvement, and UI for inference

References

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